

SQL Comprehensive Practice Question

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Database Context:

You are managing a **Library Management System**. Below are the tables you need to create:

Tables & Structure:

1. Books

- BookID (INT, Primary Key)
- Title (VARCHAR(100))
- Author (VARCHAR(100))
- Genre (VARCHAR(50))
- PublishedYear (INT)
- Price (DECIMAL(6,2))

2. Members

- MemberID (INT, Primary Key)
- FullName (VARCHAR(100))
- Address (VARCHAR(100))
- Gender (VARCHAR(10))
- JoinDate (DATE)

3. Loans

- LoanID (INT, Primary Key)
- BookID (INT, Foreign Key referencing Books(BookID))
- MemberID (INT, Foreign Key referencing Members(MemberID))
- LoanDate (DATE)
- ReturnDate (DATE)

4. Staff

- StaffID (INT, Primary Key)
- Name (VARCHAR(100))
- Position (VARCHAR(50))
- Salary (DECIMAL(7,2))
- Branch (VARCHAR(50))

Practice Questions:

1. Basic SQL Operations

1. Create the **Books**, **Members**, **Loans**, and **Staff** tables with appropriate constraints.
2. Insert **5 records** into each table.
3. Display all books authored by "**J.K. Rowling**".
4. Display all members who joined in the year **2023**.

2. Aggregate Functions & Grouping

5. Find the **total number of books** in each genre.
6. Calculate the **average price** of books published after **2015**.
7. Count the **total number of male and female members**.
8. Calculate the **total salary** of staff members working in the "**Downtown**" branch.
9. Find the **total number of loans** issued per year.

3. Sorting & Ranking

10. Display the **top 3 most expensive books**.
11. Find the **second highest salary** among staff.
12. List all books **sorted by title**.
13. Sort all members **by JoinDate** in descending order.

4. Joins & Relationships

14. Display the **titles of books borrowed by "John Doe"**.
15. Find the **names of staff** who work in the same branch as "**Downtown**".
16. Retrieve the **full details of members who have borrowed 'Harry Potter'**.
17. Display the **books borrowed by members** who joined in **2023**.

5. Subqueries

18. Find the **highest salary among staff** in the "**Uptown**" branch.
19. Retrieve the **number of books** that have never been borrowed.
20. Find the **genre of books** that have the **highest average price**.
21. List **members who have borrowed more books than the average number**.

6. Data Modification

22. Delete all books published **before 2000**.
23. Update the **salary of staff** in the "Downtown" branch by **10%**.
24. Mark all books borrowed by '**John Doe**' as returned by setting **ReturnDate** to today.
25. Add **5%** to the price of all books in the "**Fantasy**" genre.

7. Advanced SQL Concepts

26. Create a **view** showing the **books borrowed in 2024** along with borrower names.
27. Create a **stored procedure** that accepts a **MemberID** and returns all the books they have borrowed.
28. Create a **trigger** to automatically update the **ReturnDate** to today when a book is returned.

8. Bonus Challenges

29. Write a query to find **members who have borrowed at least 3 different genres** of books.
30. Calculate the **total revenue generated** from loans (assume each loan generates revenue equal to the book's price).

